Samia Soleimani

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EDUCATION

BSc Computer Engineering, First Class Honours, University of Ghana, Accra 2022 – 2026

- CGPA 3.71 / 4.00 (top 5% of class). Language of instruction: English.
- **Thesis:** *SignSpeak: A Two-Glove Fingerspelling and Word-Sign System for GhSL Translation*. Supervisor: Dr. Nii Longdon Sowah.

RESEARCH EXPERIENCE

SignSpeak: two-glove sign-to-speech system, undergraduate thesis (team of two) 2025 – 2026

- Built two multisensor gloves (ESP32, five flex sensors and an MPU-6050 IMU per glove, hand-made contact pads) that turn Ghanaian Sign Language fingerspelling and clinical word signs (FEVER, DIZZY, DOCTOR) into speech, for about US$135 in parts. On-hand sensing works in any lighting and records no images of the patient.
- Raised letter accuracy from 90% to 99% by adding contact pads to the gloves to separate A/T and U/V, letters with very similar hand shapes.
- Wrote firmware streaming 24 synchronised channels at 50 Hz with median filtering and per-session calibration, and paired frames from two gloves with independent clocks (median gap 17.6 ms, 0.33% dropped).
- Collected 1,864 repetitions from two hearing signers (26 letters, 2 control signals and 11 word signs). Shrinkage LDA reached 99.2% on letters and matched a BiLSTM on word signs while running about 1,500 times faster on a laptop (0.13 ms per window). Found and fixed a data-leakage bug in which a frozen IMU inflated accuracy.
- Tested the live gloves on three new signers who were not in the training data (about 83% on letters and 92% on word signs).
- Connected live recognition to an LLM sentence builder and text-to-speech (about 2 s from gesture to audio), and built a webcam baseline (OpenCV, MediaPipe) for a paired comparison with the gloves, now being written up.

MANUSCRIPTS IN PREPARATION

S. Soleimani, I. Bhardwaj, N. L. Sowah. “SignSpeak: A Low-Cost Sensor Glove for Fingerspelling and Clinical Sign Recognition, with a Paired Camera Baseline.” *Manuscript in preparation, 2026.*

AWARDS AND DISTINCTIONS

Dean’s Honour List, University of Ghana (three consecutive years)

1st place in Ghana, IEEEXtreme 19.0 Programming Competition, IEEE Oct 2025

- 24-hour global programming competition with over 19,000 participants.

Best Prototype, QoL Hackathon, United Nations and University of Waterloo Nov 2025

Funded Delegate, IEEE Region 8 SYP Congress, Poland (only delegate selected from Ghana) Jul 2026

TEACHING EXPERIENCE

Teaching Assistant, Computer Vision and Robotics, University of Ghana Sept 2026 – present

- Support tutorials and lab sessions for undergraduate computer vision and robotics courses.

Robotics Instructor (part-time), SES STEM Academy Jan 2025 – Sept 2025

- Taught robotics to primary school students, and coached the student team that won 3rd place in the RISE competition.

PROJECTS

Wildfire Detection System, team lead Aug 2025 – Sept 2025

- Led a team building a wildfire-risk model from NASA satellite imagery and environmental data.

Hazard Watch: hazard handling in a five-stage pipelined processor, team lead Aug 2026

- Added forwarding, load-use hazard detection and early branch resolution; 47 to 25 cycles on a test workload (1.88x).

PROFESSIONAL EXPERIENCE

AI Engineering Intern, ACCEDE Africa — Nov 2025 – Dec 2025

- Built a RAG research assistant for clinicians that answers from medical papers, cites the source behind every claim, and indexes new papers automatically when retrieval falls short (Semantic Scholar API, Azure OpenAI, ChromaDB).

Research Intern, School of Engineering, University of Ghana — Feb 2024 – Apr 2024

- Ran an experimental study comparing how efficiently different algorithms performed across a range of input sizes, and analysed the results to identify the performance trade-offs between them.

TECHNICAL SKILLS

Wearable Sensing & Signal Processing	Flex sensors, IMUs (accelerometers and gyroscopes), multichannel data acquisition, multi-device synchronisation, sensor calibration, signal filtering, sensor fusion, time-series analysis, gesture recognition
Embedded Systems & Hardware	ESP32 firmware (C/C++), Arduino, I^2C, UART, real-time data acquisition, serial communication, hardware prototyping, SolidWorks
Machine Learning	LDA, SVM, Random Forest, Logistic Regression, CNNs, BiLSTMs, feature engineering, model selection; scikit-learn, PyTorch, TensorFlow/Keras
Evaluation & Research Methods	Cross-validation, held-out and new-user testing, ablation studies, confusion matrix analysis, inference latency benchmarking, experimental protocol design, dataset collection and annotation, research paper writing
Computer Vision & AI Systems	MediaPipe hand landmark tracking, OpenCV image processing; Retrieval-Augmented Generation (RAG), LLM integration, Azure OpenAI, Gemini API, ChromaDB
Programming & Computer Architecture	Python, C/C++, MATLAB, SQL, JavaScript/TypeScript, NumPy, pandas, Git; pipelining and hazard handling, cycle-accurate simulation in C, Logisim Evolution

CERTIFICATIONS

Google UX Design Professional Certificate, Google (Coursera)	May 2024
Data Literacy Professional, DataCamp	Sep 2024
Google AI Essentials, Google	Nov 2024
Process Data from Dirty to Clean, Google	Dec 2024

LANGUAGES

Persian (native) **English** (full professional proficiency, BSc taught in English) **French** (beginner, A1, currently learning)

LEADERSHIP AND SERVICE

Aspire Leaders Program, Aspire Institute	Dec 2025
Member, IEEE SAC Region 8 Student Units Coordination Subcommittee	Jan 2026 – present
Organising Committee Member, IEEE Global Career Fair	Jul 2026 – present
Chair, IEEE Women in Engineering Student Branch, University of Ghana	Jan 2026 – Sept 2026
President and Founder, Computer Engineering Students Association, University of Ghana	May 2025 – Sept 2026
General Secretary, ARM Students Club, University of Ghana	Nov 2024 – Oct 2025

REFERENCES

Dr. Nii Longdon Sowah, Department of Computer Engineering, University of Ghana, Accra — Thesis supervisor

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